CHEN, ZHAOXI

Tsinghua University
Department of Automation

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EDUCATION

- **Tsinghua University** 08/2017 – 07/2021(Expected)
  - Bachelor of Engineering, Department of Automation
  - Accumulative **GPA**: 3.73/4.0 (3.86/4.0 for the junior year)
  - **Awards**: Friends of Tsinghua - Guangzhou Pharmaceuticals Scholarship, Academic Excellence Scholarship
  - **Courses:** Digital Image Processing, Artificial Intelligence, Computer Networks, Data Structure, Automatic Control Theory, Numerical Analysis and Algorithms, Stochastic Processes, Signal and System Analysis, Analog Electronic Technology, Digital Electronic Technology, Complex Variable Functions, Circuit Principles, etc.

PUBLICATION

- **Zhaoxi Chen***, Xiang Zhang*, Yuwang Wang, Zhen Fan, Yipeng Li, Xiu Li, Qionghai Dai. *FlyingCap: Multi-Agent Deep Reinforcement Learning for Active 3D Human Pose Estimation.* Under review for CVPR 2021.
- Zhuolin Yang*, **Zhaoxi Chen**, Tiffany Cai, Xinyun Chen, Bo Li, Yuandong Tian*. *Understanding Robustness in Teacher-Student Setting: A New Perspective.* Under review for AISTATS 2021.
- Zhong-Han Niu, **Zhaoxi Chen**, Linyi Li, Yubin Yang, Bo Li, Jinfeng Yi. *On the Limitations of Denoising Strategies as Adversarial Defenses.* Under review for CVPR 2021.

ACADEMIC EXPERIENCE

- **Secure Learning Lab (SL2), Computer Science Department, UIUC** 06/2020 – 11/2020
  Intern advised by Prof. Bo Li

  Understanding Robustness in Teacher-Student Setting: A New Perspective

  - Leveraged the teacher-student framework to understand adversarial examples of models.
  - Proposed principles about the relationship between normalized correlation (indicator of student specialization), robustness and the cause of adversarial samples.
  - Conducted experiments extensively to verify our analysis in different scenarios, including standard training, adversarial training, training with robust feature, etc.
  - Completed a research paper as the second author and submitted to AISTATS 2021.

  On the Limitations of Denoising Strategies as Adversarial Defenses

  - Proposed a symmetric transformation framework (*F-IF Framework*) to understand existing denoising-based adversarial defenses.
  - Conducted experiments extensively to reveal the limitations of denoising strategies.
  - Completed a research paper as the second author and submitted to CVPR 2021.

- **ModLab, GRASP, University of Pennsylvania** 05/2020 – 09/2020
  Intern advised by Prof. Mark Yim

  Active Reconfiguration with SMORES-EP, a Modular Robot

- Proposed a graph neural network(GNN) based framework for configuration identification of modular robots consist of a variable number of modules.
- Addressed the problem of lack of training data from a perspective of few-shot learning.
- Developed an *Unity* based simulator for modular robot, SMORES-EP, where user can construct, verify and simulate their configuration designs of modular robots.

- **Microsoft Research Asia** 03/2020 – 10/2020
  Research Assistant advised by Dr. Yuwang Wang

  Multi-Agent Deep Reinforcement Learning for Active 3D Human Pose Estimation

  - Became the first to address active 3D pose estimation problem with a distributed multi-agent based deep reinforcement learning framework.
  - Applied kinematic constraint and inter-agent communication to maintain realistic camera motions.
  - Added value prediction module to policy network to predict future movements of cameras.
  - Illustrated the feasibility of the system in a simulation environment.
  - Generated a research paper as the first author and submitted to CVPR 2021.

- **BBNC Lab, Department of Automation, Tsinghua** 08/2018 – 09/2020
  Research Assistant advised by Prof. Yipeng Li

  "Wild Array" Collaborative UAV Intelligent System

  - Led a team of nine to develop a collaborative intelligent UAV system consists of 3 UAVs.
  - Constructed the system from scratch, including body design, system hardware architecture and multi-sensor fusion.
  - Tested algorithms including autonomous positioning, 3D mapping, human body detection and swarm path planning with the system.
  - Optimized the visual-inertial odometry VINS-Mono on TX2 for real-time use.
  - Applied the system into Mohamed Bin Zayed International Robotics Challenge 2020.

PATENT

- **CN110296702A** – A Pose And Motion Estimation Method For Tight Coupling Of Visual Sensor And Inertial Navigation
  - Patent approved; the technique could avoid the shortcomings of the single sensor and provide a robust and efficient system method for SLAM system application in different scenarios.

ACTIVITIES

- **The 17th Tsinghua Siyuan Excellent Student Training Program** 05/2018 – Present
  - Selected 35 out of 3300 students.
  - Training in leadership skills and social responsibility.
  - Provided volunteer education for Shangluo, Shaanxi, China.
- **Saxophone Performer – Tsinghua University Student Art Troupe** 09/2017 – Present

ENGLISH PROFICIENCY AND OTHER SKILLS

- **English Proficiency:** TOEFL 107 (MyBest); GRE 157+169+3
- **Programming Skills:** Proficient in C/C++, Python, Java, C#, MATLAB, Verilog, LaTeX
- **Software & System:** Linux, ROS, Git, Unity, OpenCV, PyTorch, Numpy, Altium Designer